**Ezequiel Boero <ezequiel.boero@adagioinstitute.org>**

# Re: Review #3 Convex Perturbation Paper

**Ezequiel Boero** <ezequiel.boero@adagioinstitute.org> Wed, Sep 23, 2026 at 7:34 PM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Ben,

I'm fulfilling here my reply to your responses on paper 3.
So far I could see, Sara and I raised three principal issues; all three are addressed.

**What I verified.** The identity (6) to machine precision on 4,000 random instances in dimensions 2 to 6, without symmetry or commutativity;
Proposition 3.2(i) as an identity (maximum deviation $4.4\cdot10^{-16}$) and (ii) convexity, the Lipschitz constant M*, S(0) = D_CK, and the linear envelope
with no violations over 600 multi-triple families;
Theorem 4.3(i)–(iii) over 3,000 random parameter sets, including root selection, strict monotonicity of E on $[g_1,1]$, and the chain $g_1 \le g_0$;
the new quadratic constant η(ε) of Proposition 4.8, discriminant included, and its continuity at M = 0; the Minkowski form of Theorem 5.1 on
300 families, where it is never weaker than the factor-of-two form;
Proposition 6.1's bound on 1,500 random state-dependent weights; the Bayes weight of Remark 6.2 against direct conditioning on 2,000
kernel-and-law pairs ($2.2\cdot10^{-16}$); and every number in Section 4.3 in both regimes, including the interior zero at 0.360328, the rebound peak
0.022712 near g = 0.6802, and $g_1 = 0.80709$ against $g_0 = 0.83445$.
I also audited the internal numbering after the deletions of Remarks 4.6 and 4.11 and found no dangling reference.

Proposition 3.2 is answering to my Finding 2: it replaces a false uniqueness inference with a characterization, which is more than the correction required.
Remark 6.2 is the right answer to Finding 3 for the same reason (you computed the weight instead of withdrawing the claim, and that computation
is the substance of the objection). I note also that you carried Sara's quadratic-envelope device over to Proposition 4.8 on your own,
with the correct discriminant $(m_0 - M)^2 + 4M\varepsilon$.

Two results are available that would strengthen the paper. Neither is a correction and neither conditions my recommendation.
I set them out because in both cases the paper states something it could prove.

**1. Theorem 4.3 is sharp, not merely sign-blind.** Remark 4.5 explains the factor-of-eight gap as the price of summarizing the
pencil by two numbers, and says S cannot be sharpened. Correct, but that is an explanation; the corresponding statement is one line and stronger.
If at an extremal triple K = −cX with c = M*/D_CK, then ‖X − gK‖ = D_CK + gM* for every g, so D_CK(δt;Γ_g) = E(g) identically and g_term = $g_1$ exactly.
Hence no bound expressed in D_CK and M* alone can improve Theorem 4.3 (the quadratic envelope is attained).

The configuration is realized inside your own example.
With the Glauber reference at reflection weight q = ½(1 − e^{−2λ}) one has K = 2(q − 0.3)(0.2 − q)Z, which points against X = −0.04 Z
exactly when q ∈ (0.2, 0.3), i.e. λ ∈ (0.2554, 0.4581). At λ = 0.35 I get $\max_{g\in[0,1]} |D(t,s,r;\Gamma_g) - E(g)| = 3.5\cdot10^{-17}$ and
g_term = $g_1$ = 0.531092.
Added to Remark 4.5 this turns a caveat into an optimality statement and settles my Question 7: the gap in the interaction-dominant regime
is the price of alignment, and there is nothing left to recover.

**2. A scale-level instance.** The abstract says the deficit up to scale δt equals (1 − g)S(g), and that the response can vanish at one interior
coupling and can rebound. Those are claims about D_CK(δt;·). Proposition 3.2(iii) says what interior vanishing requires at that level:
X = gK at *every* triple of T(δt) with one common g (a rigid proportionality, not a coincidence at a point).

Section 4.3 exhibits the phenomenon at a single triple, and item 10 of your response says you did not construct a family.
So the possibility the abstract highlights is not instantiated at the level at which it is stated.

It is instantiated by affine extrapolation. Let Γ_a and Γ_ref be divisible and let g* ∈ (0,1) satisfy Γ_a ≥ g*Γ_ref entrywise;
set $\Gamma_0$ := (Γ_a − g*Γ_ref)/(1 − g*). Then $\Gamma_0$ is column-stochastic with $\Gamma_0(s \leftarrow s) = I$, generically indivisible, and Γ_{g*} = Γ_a is divisible at every triple.
Conversely, take Γ_a := Γ_g. So the mixture is divisible at an interior coupling if and only if the substrate is an affine extrapolation of two divisible
families through the reference (the global counterpart of Proposition 3.2(iii)'s pointwise proportionality, and the reason for the rigidity:
the substrate must lie on the line through Γ_ref and a divisible family, beyond that family).

A two-state, time-homogeneous instance: Γ_ref = Glauber(λ = 2), Γ_a = Glauber(λ = 1), g* = 0.3603.
Positivity holds for every τ ≥ 0 (minimum entry $4.4 \cdot 10^{-5}$), and $1 - 2p(\tau) = [e^{-2\tau} - g^* e^{-4\tau}]/(1 - g^*)$ is not multiplicative,
so $\Gamma_0$ is indivisible: $D_{CK}(3;\Gamma_0) = 0.055029$.
Since X = g*K at every triple (the ratio ‖X‖/‖K‖ came out 0.360300 at all 2,600 triples of a grid) the response has the closed form

$D_{CK}(\delta t;\Gamma_g) = (1 - g)|g^* - g|\, M^*(\delta t)$, $M^*(3) = 0.152731$, attained at r − s = t − r = (ln 2)/2.

At scale level this exhibits the interior zero at g* = 0.3603, the rebound to 0.015625 at g = (1 + g*)/2 = 0.68015, and, for any tolerance below that peak,
genuine separation of the two thresholds: at ε = 0.01, g_first = 0.2705 and g_term = 0.8721, with the deficit back above ε on (0.4885, 0.8721),
against $g_1 = 0.9500$ and $g_0 = 0.9519$.

I would suggest this in place of the extension discussed in item 10. The family you propose there
(the two-state family of [1, Section 6] mixed with the Glauber reference) would realize the identity and the bounds at scale level,
but neither the interior zero nor the threshold separation, which are the two facts the two-threshold framework exists to accommodate.
Three lines give you both, and they give the first-crossing/terminal distinction a family-level example rather than a pointwise one.
On that basis I do not think the pointwise example undersells the theorems; what currently outruns the evidence is the abstract's
two scale-level claims, and this closes them.

**Four points of wording.**

(i) The abstract keeps "exact vanishing is guaranteed *only* at g = 1." The intended reading (g = 1 is the only coupling at which vanishing is guaranteed)
is correct and the body says it precisely; but this is the word order that produced my Finding 2 and Sara's item 27, and a reader will take "only at g = 1"
for the uniqueness claim Proposition 3.2(iii) now disclaims.
"Guaranteed at g = 1, and possible at no more than one interior coupling" removes the ambiguity.

(ii) After Proposition 6.1, "for constant weight (δg = 0) it reduces to Theorem 3.1": with δg = 0, (18) reduces to the operator-norm bound (7),
not to the exact identity, which the inequality has already discarded.

(iii) Remark 6.2's weight is defined only at states of positive probability under the mixture; if π_s^Y(j) = π_s^Z(j) = 0 the conditional kernel
is undetermined in column j. "For states j of positive probability under the mixture" fixes it; and since Proposition 6.1 needs no regularity of g,
completing a null column with the constant weight g keeps the family in class (16) and leaves δ_g and the estimate unchanged,
which is worth the half-sentence.

(iv) In the proof of Theorem 4.3(i), the radicand is positive because $(D + M)^2 - 4M\varepsilon > (D - M)^2 \geq 0$ when ε < D; that is currently left to the reader.

Subject to those four points of wording, I regard the substantive issues of both reports as resolved or correctly delimited and I have no further
substantive revision to ask for on this paper.
On mathematical depth I agree with you and with Sara: the techniques are elementary, the paper now says so, and the contribution is the exact identity,
the pencil form, and the two-threshold structure the identity forces.
That is a real contribution, correctly positioned against Anand–Brun and De Santis–Giovannetti.

Best,
Ezequiel

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